

GRID GUARDIAN

Power Outage Prediction & Equipment Failure Advisor

Predict. Prioritize.
Prevent.

GridGuardians | U1 – Power Outage Prediction & Grid Equipment Failure Advisor



Power Failures Can Be Predicted — But Action Comes Too Late

The Risk Signals Exist

**Temperature • Vibration •
Partial Discharge • Oil
Condition**

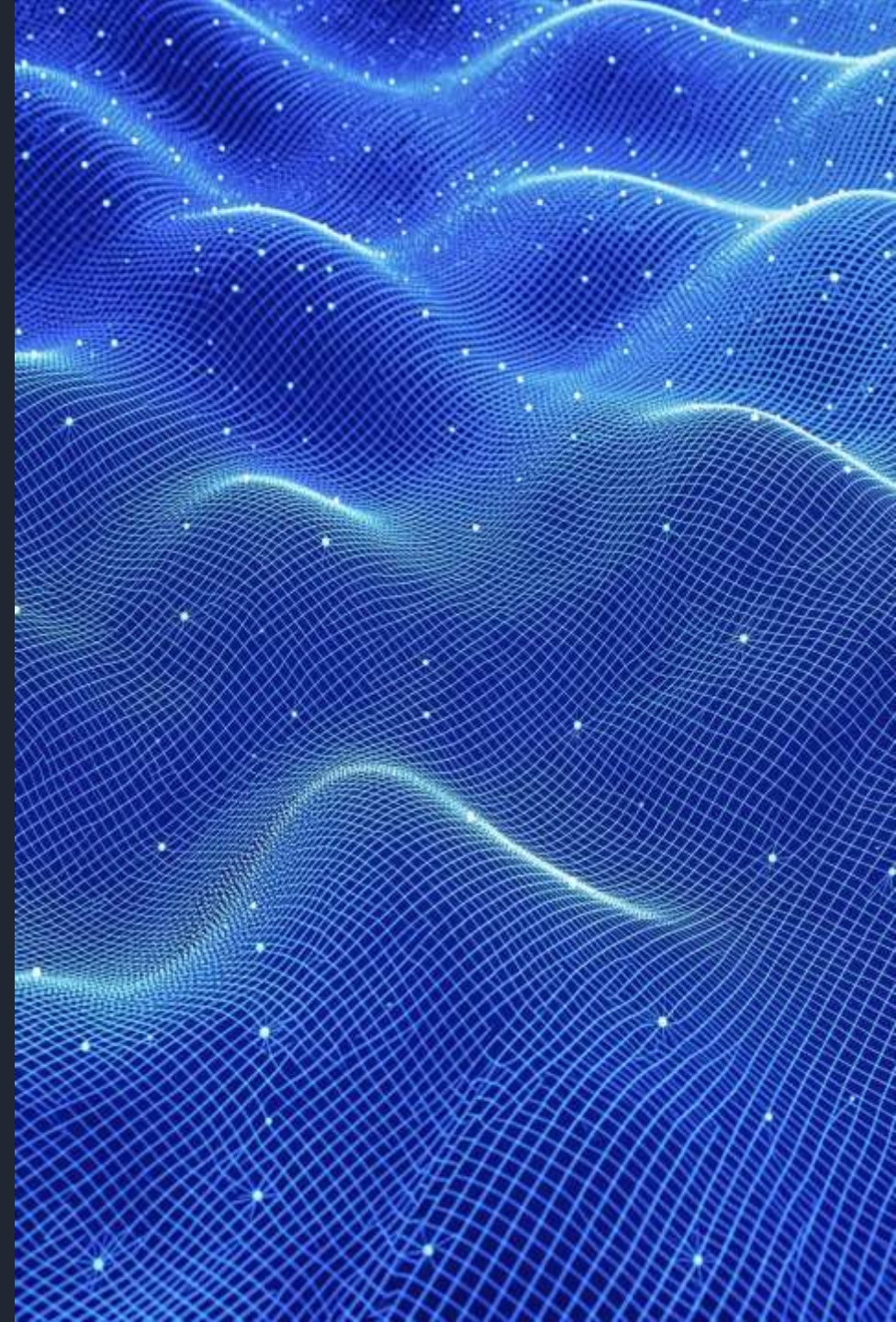
Weather • Wind, rain, storms
increase failure risk

**Historical patterns • Previous
failures** contain warning
signals

The Problem

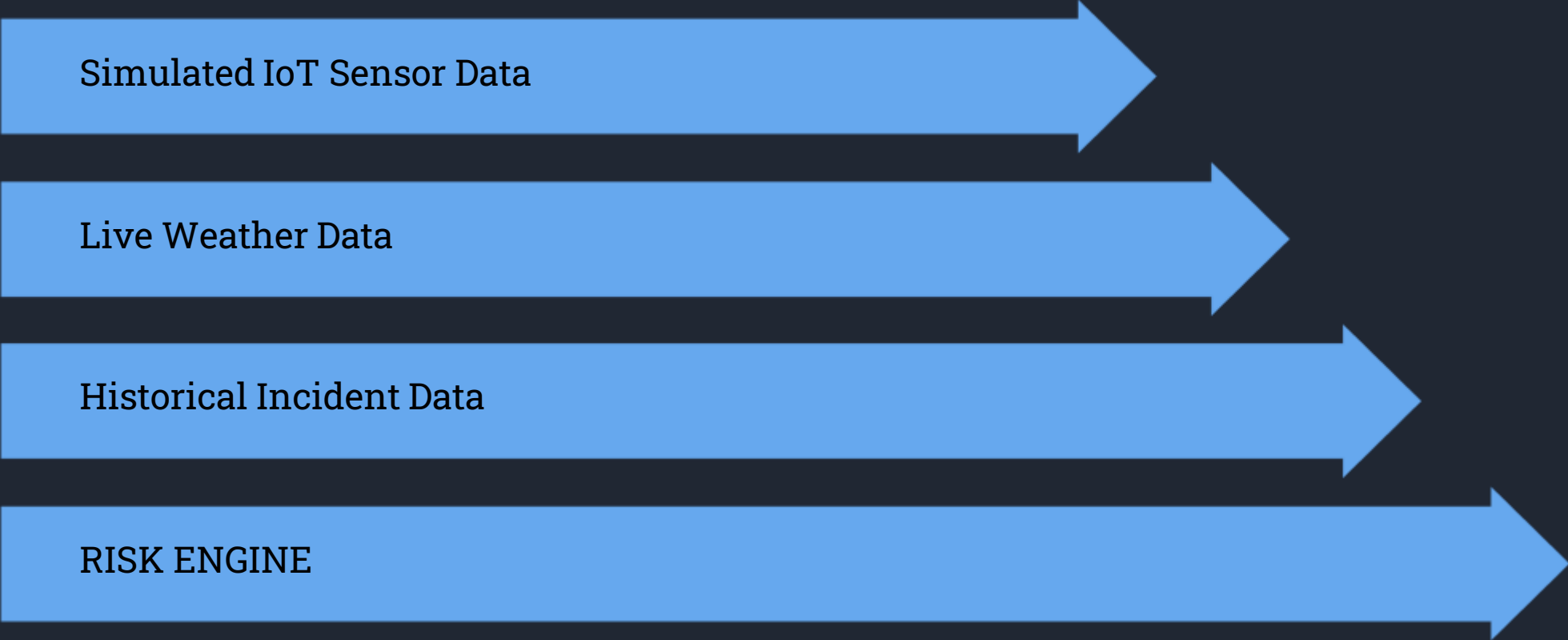
Utilities detect risk signals but lack a system to prioritize which assets matter most and act before failure occurs.

Which asset should we act on BEFORE it fails?



Grid Guardian Turns Risk Signals into Preventive Decisions

A risk intelligence engine that transforms sensor data, weather, and historical patterns into actionable maintenance and crew deployment decisions.



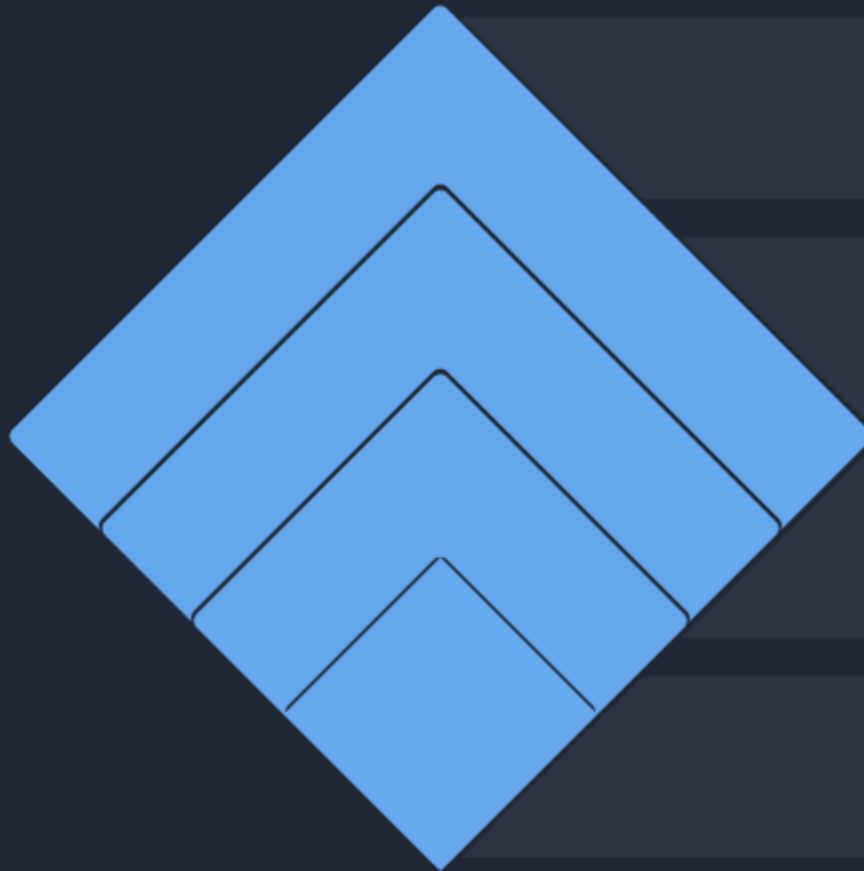
Simulated IoT Sensor Data

Live Weather Data

Historical Incident Data

RISK ENGINE

How Grid Guardian Works



Risk Calculation Engine

Sensor Health (50%) + Weather Risk (30%) + Historical Risk (20%)
Composite Risk Score: 0–100

Historical Incidents

Previous failures, weather events, outage duration

Live Weather Data

Wind, rain, temperature, storm conditions

Simulated Sensor Data

Temperature, vibration, partial discharge, oil quality

LIVE data is sourced from weather APIs. **SIMULATED** sensor inputs are used for the prototype.

Prototype uses simulated sensor data and assets. Live weather data only.

From "At Risk" to "What Should We Do?"

Detect → Prioritize → Act

CRITICAL ASSET

TX-203

87 / 100

CRITICAL
HIGH GRID IMPACT

Temperature	Elevated
Vibration	Rising
Partial Discharge	Critical
Oil Quality	Normal
Weather Risk	Storm
Historical Risk	High

SYSTEM PRIORITY RANKING

#1 SS-305 91.7 • #2 TX-203 67.3 • #3 TX-101 66.4

RECOMMENDED ACTION

Inspect transformer immediately.

Crew C-01
Transformer Specialist
Northgate
Pre-position before expected risk window

DETECT

Sensor + weather + history

PRIORITIZE

Risk score and grid impact

ACT

Crew and maintenance dispatch

IBM Bob Turns Grid Intelligence into an AI Assistant

Natural-language access to Grid Guardian's risk intelligence through **MCP (Model Context Protocol)**

Operator Questions & Answers

Q: Which asset is highest priority?

A: **SS-305** at **91.7** risk score — critical grid impact.

Q: Why is TX-203 risky?

A: High equipment risk + weather exposure + historical incidents.

Q: What maintenance should we perform?

A: Inspect transformer immediately. Pre-position **Crew C-01**.

Q: Which crew should be pre-positioned?

A: Transformer specialist crew in the Northgate zone before the risk window opens.

IBM Bob → MCP → Grid Guardian

IBM Bob

MCP Server

Grid Guardian Risk Engine



IBM Bob is not a chatbot — it is an **intelligent MCP-based assistant layer** that queries live risk intelligence and returns structured operational guidance.

MCP Tools & Functions

`get_risk_ranking()`

Returns all assets sorted by composite risk score

`get_asset_risk()`

Detailed risk breakdown for any specific asset

`get_maintenance_plan()`

Recommended inspection and repair actions

`get_crew_prepositioning_plan()`

Specialist crew assignments and deployment timing

`list_monitored_assets()`

Full inventory of assets currently under observation

What Does Success Look Like?

Prototype Metrics

8

Monitored Assets

Prototype grid assets under continuous risk scoring

0–100

Risk Scoring Range

Composite scoring across sensor, weather, and historical inputs

10 SEC

Dashboard Refresh

Near real-time risk updates for operational awareness

11

Automated Tests Passed

Full test suite passing across all system components

Operational Impact Chain

→ Earlier Risk Identification

→ Prioritized Maintenance

→ Better Crew Readiness

→ Faster Operational Decisions

Current Risk Ranking

Asset	Score	Level
SS-305	91.7	Critical
TX-203	67.3	High
TX-101	66.4	High
SS-508	57.9	Elevated

☒ LIVE

Weather data — real-time API feed

● SIMULATED

Sensor telemetry, grid assets, historical incidents, crew data



Grid Guardian moves utilities from reactive response toward proactive risk-based action.

GRIDGUARDIANS

Predict. Prioritize. Prevent.

TEAM

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Frontend & Dashboard

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Backend & Risk Engine, IBM Bob Integration

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Documentation & Presentation

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FROM PREDICTION TO PREVENTION